Simultaneous Power-Up Synchronization

1	Parameter / Node	DK1 (Central)	DK2 (Central)	DK3 (Peripheral)	DK4 (Peripheral)
	Role	Central	Central	Peripheral	Peripheral
	Wi-Fi Availability	Yes (Wi-Fi v5)	Yes (Wi-Fi v5)	No Wi-Fi	No Wi-Fi
	Power Status	OFF → ON	OFF → ON	OFF → ON	OFF → ON
	Power-Up Method	Simultaneous	Simultaneous	Simultaneous	Simultaneous
	Priority Level	Equal	Equal	N/A	N/A
	Synchronization Result	Synced with DK4	Synced with DK3	Synced with DK2	Synced with DK1
	Final State	Active & Synced	Active & Synced	Active & Synced	Active & Synced
	Test Verdict	PASS	PASS	PASS	PASS

2	Parameter / Node	DK1 (Central)	DK2 (Central)	DK3 (Peripheral)	DK4 (Peripheral)
	Role (Initial)	Central	Central	Peripheral	Peripheral
	Wi-Fi Version	v6	v5	—	—
	Priority Level	**High**	Lower	N/A	N/A
	Power Status	OFF → ON	OFF → ON	OFF → ON	OFF → ON
	Power-Up Method	Simultaneous	Simultaneous	Simultaneous	Simultaneous
	Sync Decision Basis	Highest Wi-Fi priority	Not selected	Select DK1	Select DK1
	Actual Sync Result	DK3 & DK4 synced	No peripherals synced	Synced with DK1	Synced with DK1
	State After Sync	**Remains Central**	**Remains Central (Standby)**	**Switched to Jupiter mode**	**Remained Peripheral (BUG)**
	Expected State	Central	Central (Standby)	Repeater mode	**Repeater mode**
	Test Verdict	PASS	PASS	PASS	FAIL

3	Staggered Power-Up (DK4 After Initial Sync)				
	Parameter / Node	**DK1 (Central, v6)**	**DK2 (Central, v5)**	**DK3 (Peripheral → Repeater)**	**DK4 (Peripheral)**
	Role (Initial)	Central	Central	Peripheral	Peripheral
	Wi-Fi Version / Priority	v6 / High	v5 / Lower	—	—
	Power-Up Timing	T0 (Simultaneous)	T0 (Simultaneous)	T0 (Simultaneous)	**T1 (After sync established)**
	Initial Sync Outcome	Selected as main parent	Not selected	**Synced with DK1 (Periodic Sync established)**	Not powered yet
	Role After Initial Sync	Remains Central	Remains Central (Standby)	**Becomes Repeater (synced to DK1)**	Powers up later

	DK4 Sync Target	—	—	—	**Synced with DK3 (Repeater)**
	Final Topology	Parent Central	Backup Central	Repeater linked to DK1	Peripheral linked to DK3
	Test Verdict	PASS	PASS	PASS	PASS

4	**Test Table 4A – Central First, Peripheral Later (Delay 10–30s)**				
	Parameter / Node	DK1 (Central)	DK2 (Peripheral → Repeater)		
	Role (Initial)	Central	Peripheral		
	Wi-Fi Version	v5	None		
	Power-Up Timing	T0 (first)	T0 + 10–30s		
	Sync Expectation	Accept link	Discover + sync to DK1		
	Actual Result	DK2 successfully linked	Synced with DK1 successfully		
	State After Sync	Remains Central	Switches to Repeater		
	Test Verdict	PASS	PASS		

5	**Test Table 4B – Peripheral First, Central Later (Peripheral powers up ≥30s earlier)**				
	Parameter / Node	**DK1 (Central)**	**DK2 (Peripheral → Repeater)**		
	Role (Initial)	Central	Peripheral		
	Wi-Fi Version	v5	None		
	Power-Up Timing	T0 + ≥30s (second)	T0 (first)		
	Sync Expectation	Start later, still accept link	Wait/advertise/scan until DK1 appears, then sync		
	Actual Result	Successfully established link with DK2	Successfully synced after DK1 powered up		
	State After Sync	Remains Central	Switches to Repeater		
	Repeated Trials	Works consistently (many runs)	Works consistently (many runs)		
	Test Verdict	PASS	PASS		

6	**Peripherals First, Central Later (DK2 Missing)**				
	Parameter / Node	**DK1 (Central)**	**DK3 (Peripheral)**	**DK4 (Peripheral)**	
	Role (Initial)	Central	Peripheral	Peripheral	
	Wi-Fi Version	v6	—	—	
	Priority Level	High	N/A	N/A	

Power Status	OFF → ON	OFF → ON	OFF → ON
Power-Up Timing	T0 + ~30s	T0 (first)	T0 (first)
Power-Up Method	Delayed	Peripherals first	Peripherals first
Sync Decision Basis	Only available central	Select DK1 when it appears	Select DK1 when it appears
Actual Sync Result	DK3 & DK4 synced	Synced with DK1	Synced with DK1
State After Sync	Remains Central	Switched to Jupiter mode (if it did)	Remained Peripheral (BUG) (if it did)
Expected State	Central	Repeater mode	Repeater mode
Test Verdict	PASS	PASS	FAIL (if DK4 stayed peripheral again)